Implement Trie (Prefix Tree) (medium)

Problem Statement

Design and implement a Trie (also known as a Prefix Tree). A trie is a tree-like data structure that stores a dynamic set of strings, and is particularly useful for searching for words with a given prefix.

Implement the `Solution` class:

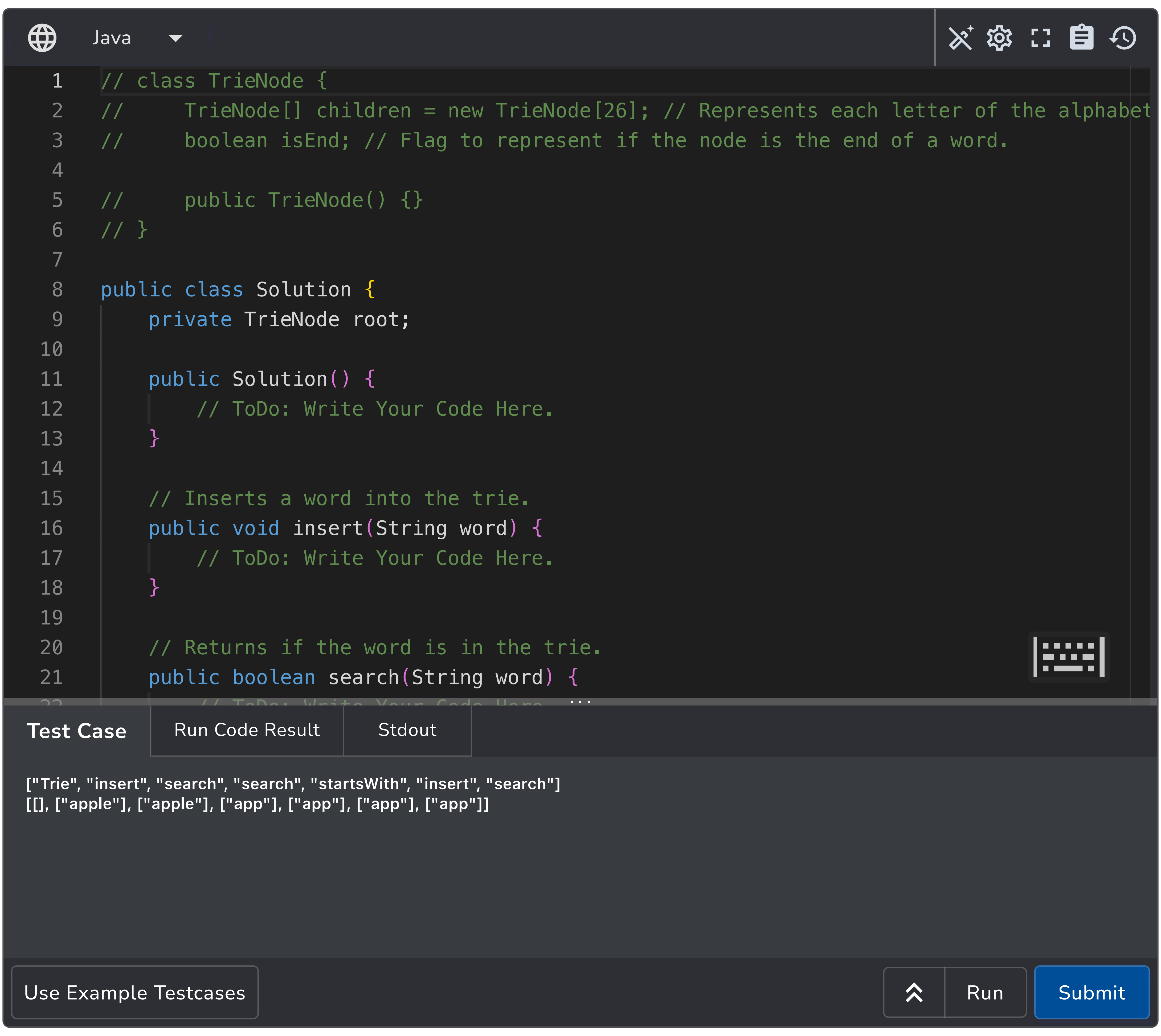
- `Solution()` Initializes the object.
- `void insert(word)` Inserts `word` into the trie, making it available for future searches.
- `bool search(word)` Checks if the word exists in the trie.
- `bool startsWith(word)` Checks if any word in the trie starts with the given prefix.

Examples

1. **Example 1:**
   - **Input**:
     - Trie operations: `["Trie", "insert", "search", "startsWith"]`
     - Arguments: `[[], ["apple"], ["apple"], ["app"]]`
   - **Expected Output**: `[-1, -1, 1, 1]`
   - **Justification**: After inserting "apple", "apple" exists in the Trie. There is also a word that starts with "app", which is "apple".
2. **Example 2:**
   - **Input**:
     - Trie operations: `["Trie", "insert", "search", "startsWith", "search"]`
     - Arguments: `[[], ["banana"], ["apple"], ["ban"], ["banana"]]`
   - **Expected Output**: `[-1, -1, 0, 1, 1]`
   - **Justification**: After inserting "banana", "apple" does not exist in the Trie but a word that starts with "ban", which is "banana", does exist.
3. **Example 3:**
   - **Input**:
     - Trie operations: `["Trie", "insert", "search", "search", "startsWith"]`
     - Arguments: `[[], ["grape"], ["grape"], ["grap"], ["gr"]]`
   - **Expected Output**: `[-1, -1, 1, 1, 1]`
   - **Justification**: After inserting "grape", "grape" exists in the Trie. There are words that start with "grap" and "gr", which is "grape".

Try it yourself

Try solving this question here:



```java
// class TrieNode {
//     TrieNode[] children = new TrieNode[26]; // Represents each letter of the alphabet
//     boolean isEnd; // Flag to represent if the node is the end of a word.

//     public TrieNode() {}
// }

public class Solution {
    private TrieNode root;

    public Solution() {
        // ToDo: Write Your Code Here.
    }

    // Inserts a word into the trie.
    public void insert(String word) {
        // ToDo: Write Your Code Here.
    }

    // Returns if the word is in the trie.
    public boolean search(String word) {
```

```
["Trie", "insert", "search", "search", "startsWith", "insert", "search"]
[[], ["apple"], ["apple"], ["app"], ["app"], ["app"], ["app"]]
```

← Previous
Solution: Graph Valid Tree

Next →
Solution: Implement Trie (Prefix Tree)

☑ Mark as Completed